



MINING DEALER BEST PRACTICE

Longwall Move Box Process

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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**June 2016
1605-1.07-1437**

1.0 Introduction

Description:

Providing prepackaged boxes with critical spares for Longwall moves greatly enhances customer satisfaction by preventing delays during the move period. Equipment can quickly be put back into service reducing the downtime and subsequent cost to the customer.

Scope:

This Best Practice will describe a process that will enhance a dealer support strategy to be implemented at any Underground Longwall operation.

History:

The move box process pre-dates the Caterpillar Inc. acquisition of the EMP products. In the underground mining business, the delays brought about by having to travel a long way to get the necessary parts to keep production equipment running can be more challenging than surface mining. This process has been in existence to help mitigate those challenges for the last 15 years.

Impact:

Having immediate access to all critical spares identified previously allows maintenance personnel to quickly react to situations that could have jeopardized the Longwall move schedule and subsequent production loss due to missed start up dates. It is an opportunity for a dealer to work closely with their customer and improves the overall customer/dealer relationship.

Resource Requirements:

This process requires minimal capital expense to initiate and in essence is an extension of the warehouse management that already exists at all dealerships.

2.0 Implementation Steps

Target Customer:

Once a target customer has been identified, research should be done or at least confirmed on the specification of the equipment that customer is operating.

Critical Parts Listing:

Service personnel should then create the lists of parts based on their experience and product knowledge. These lists should cover all utility vehicles and shield haulers as well as parts likely to be damaged on the Longwall equipment being relocated.

Training:

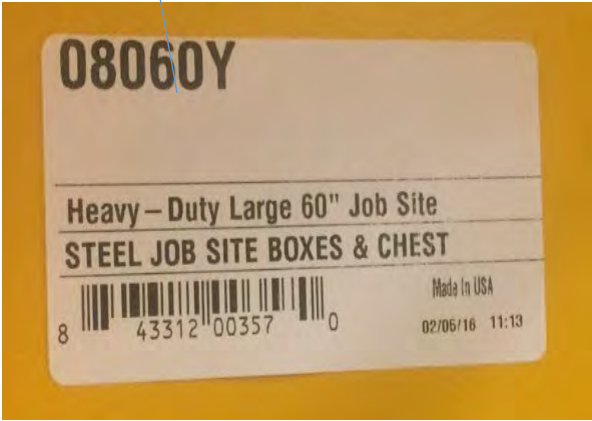
Employees who are involved in this process need to be trained on the process itself and aware of their role and responsibilities in order to ensure a successful implementation.

Equipment:

Once it has been determined how many boxes may be required, a purchase order request needs to be submitted, approved and then the purchase made. The boxes used by CB Mining are typically around \$700 US. (see addendum 1)

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Addendum 1

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3.0 Best Practice Description

Pre-delivery steps:

Check warehouse stock of the previously created critical parts lists. Create a quote from those CPL's. A dealer representative should then set up a meeting with the target customer and present the program and the CPL proposals. Any requirements from the customer over and above should be added to the CPL's and re-submitted to the customer. Establish the protocol on additional larger items such as electric motors, axles and winches as to whether the dealer keeps them locally or in the mine warehouse.

This should be done 4-6 weeks before any Longwall move is scheduled as they have to approve the parts quote and allow time for the logistics.

Delivery:

Once a PO has been received the production department sets up a work order to which the warehouse can then issue the quoted parts. An inventory contents list needs to be created for each move box being supplied and affixed externally and a copy enclosed within the box itself. The boxes are then locked and delivered to the mine. Upon receipt at the mine warehouse a dealer representative should schedule a visit and handover the keys to the boxes once the contents have been inventoried and the contents approved by the customer. The boxes can then be transported underground to a convenient location for the move.

Post-delivery:

When the Longwall move has been completed the move boxes are transported back out to the mine warehouse. The contents are inventoried and the contents approved by both a dealer representative and the customer before being returned to the dealer warehouse. The keys are handed back to the dealer representative at this time and the boxes shipped back to the dealership. The results of this exit inventory determines what the customer will be invoiced

Qualifiers:

If no preference is given by the customer with regard to new or exchange components then the dealer retains that discretionary right of supply.

1. When a new part is supplied no used core will be accepted in return.
2. Any exchange part core not returned will be invoiced at the new retail price.
3. Move boxes will be provided for a period not to exceed 4 weeks. If this time period is exceeded the box contents will be invoiced for the full value of the quote.
4. All the parts will be quoted at the normal customer list price including applicable discount.
5. The customer PO has to be a minimum deposit of \$1,500.00 US per box. The deposit is required as sometimes the customer will damage the box (see addendum 2). The deposit amount can vary depending on the costs by the dealer to produce the box.
6. Any maintenance or replacement of boxes will be deducted from the deposit
7. No parts can be returned after the exit inventory and boxes have been returned to the dealership

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Addendum 2

4.0 Benefits

The intent of the program is to immediately make available parts which may be required during a Longwall move, both on the Longwall itself, or the equipment being utilized to facilitate the move. This reduces the downtime and helps the customer achieve his production startup target date. This availability of parts increases opportunity of the dealer capturing the part sale with an OEM part.

The implementation of this program enhances both the reputation of the dealer and the customer satisfaction and loyalty

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5.0 Supporting Attachments/References

Typical move box contents as used by CB Mining

Standard SH620 Move Box (New Style DM Scoop)		
SAP Part Number	Description	QTY
317666-SR	Logic Card (Contactorless Panel)	1
262258-SR	IGBT Tram Panel (Contactorless Panel)	1
262276-SR	IGBT Pump Starter	1
898204-SR	Assy LH Control Handle	1
349512-SR	Bell Crank Lift Cylinder	1
606311-SR	Steering Cylinder	1
224431-SR	Tilt Cylinder	1
606310-SR	Battery Lift Cylinder	1
219380	Push/Pull Cable (Breaker Reset)	1
226391	1/2" X 1.25" SS Bolt (Elec Enclosures)	8
217548	1/2" SS FLAT WASHER (Elec Enclosures)	8
229691	1/2" SS LOC WASHER (Elec Enclosures)	8
391289	Tape Switch 6"	1
391290	Tape Switch 12"	1
236465	Accelerator Switch	1
212568	Current Transducer	1
606982	Fuse 30A	4
229108	Pump Fuse 125A	2
215155	Tram Fuse 450A	2
317144	Diode	4
362229	Relay Standard	2
358294	Relay Socket Standard	2
625570	I.S. Relay	1
391063	Solenoid	2
248709	Head Light	1
222316	Head Light	1
272244	Main Circuit Breaker 800A	1
222586	UVR Coil	1
238231	U Joint	2
606536	Drive Line Front and Rear	1
244427	Drive Line Yoke (Reducer)	1
351727	Drive Line Yoke Axle	1
476-1268	7/8"X12" Bolt Axle Mounting	8
222177	7/8" Loc Nut Axle Mounting	8
207380	Wheel Mounting Washer	16
126319	Wheel Mounting Bolt	16
205542	Winch (Bolt In)	1
234784	2 Speed Winch Motor	1

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276923	Connecting Link Winch Chain	1
332439	Winch Drive Chain	160
332351	Winch Drive Sprocket	2
219882	Steer Cyl Nut	2
208878	Steer Cyl Washer	2
235848	Steer Cyl Pin	2
258729	Brake Hand Pump	1
244413	2 Speed Winch Valve	1
244374	Main Relief	1
305179	Unloading Valve	1
357246	Priority Flow Control Valve	1
373567	Rev Brake Mod Valve	1
271749	Main Gear Pump	1
248304	Solenoid Valve	1

Standard SH630 (636) MOVE BOX		
SAP Part Number	Description	QTY
317666-SR	Logic Card Contactorless	1
253363-SR	Logic Card Contactor	1
253741-SR	IGBT Tram Panel Contactor	1
262258-SR	IGBT Tram Panel Contactorless	1
262276-SR	IGBT Pump Starter	1
208147-SR	Bell Crank Lift Cyl	1
241860-SR	Steering Cylinder	1
224431-SR	Tilt Cylinder 8.5" 4 Port	1
235793-SR	Tilt Cylinder 8.5"2 Port	1
219380	Push/Pull Cable	1
236465	Accelerator Switch	1
212568	Current Transducer	1
248734	Light Switch	1
213178	Master Switch	1
229108	Pump Fuse 125A	2
621522	Tram Fuse 500A	2
317144	Diode	4
237497	Relay Standard	1
251044	Relay Socket Standard	1
625570	I.S. Relay	1
391063	Solenoid	1
222316	Head Light	1
272244	Main Circuit Breaker 800A	1
222586	UVR Coil	1

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238231	U Joint	1
288434	Drive Line Frt Small Axle	1
300892	Drive Line Rear Small Axle	1
285137	Drive Line Rear 8.5C	1
285134	Drive Line Front 8.5C	1
282192	7/8"X12" NF Small Axle Mounting	8
230844	7/8" NF Nut Small Axle Mounting	8
226800	Small Axle Wheel Stud	16
226800	Small Axle Wheel Nut	16
245162	Small Axle Wheel Nut Cap	16
316966	1.25"X12.5" Big Axle Mounting	8
236002	1.25" Flex Loc Nut Big Axle Mounting	8
322845	Axle Tech Wheel Stud	16
241586	Axle Tech Wheel Nut	16
220975	Spring Brake	1
205542	Winch	1
234784	2 Speed Winch Motor	1
207408	Winch Coupling	1
220439	Winch Adapter	1
218087	Castle Nut 1 1/4"	2
230938	Steer Cyl Pin 828	2
244413	2 Speed Winch Valve	1
244374	Main Relief	1
227628	Charging Valve	1
263747	Brake Dif Loc Valve	1
207847	Gear Pump	1
248304	Solenoid Valve	1
246445	Canopy Post	1
205893	Canopy Mount	1

Standard SH650 DC MOVE BOX		
SAP Part Number	Description	QTY
317666-SR	Logic Card Contactorless	1
262258-SR	IGBT Tram Panel Contactorless	1
262276-SR	IGBT Pump Starter	1
898609-SR	Control Handle	1
317045-SR	Lift Cylinder	1
317043-SR	Tilt Cylinder	1
236465	Accelerator Switch	1
237497	Relay Standard	1
251044	Relay Socket Standard	1
625570	I.S. Relay	1

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391063	Solenoid	1
222316	Head Light	1
212568	Current Transducer	1
253392	Driver Board	1
333365	128-12 DC-DC Converter	1
214407	128-24 DC-DC Converter	1
322182	Circuit Breaker 800Amp	1
222586	UVR Coil	1
240361	Resistor 33K	3
317144	Diode	4
317145	PTC Thermister	1
229108	Fuse 125 Amp	2
215705	Fuse 500 Amp	2
244022	Modular Cable Phone Cord	1
240034	LCD Display	1
320868	Drive Line Front and Rear	1
323192	U Joint Front and Rear	2
284718	Speed Reducer Yoke Front and Rear	1
326298	Axle Yoke Front and Rear	1
322558	1.25"X10" Axle Mounting Bolt	8
236002	1.25" Loc Nut Axle Mounting	8
320716	Wheel Mounting Nut	16
373059	Adaptor Winch Motor	1
328816	Winch Motor	1
328650	Rev Modulating Brake Valve	1
305179	Unloading Valve	1
224929	Accumulator	1
318417	HRC Valve	2
365543	Trunion HRC Valve	1
326643	Steering Adapter HRC Valve	1

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Marshall County Coal Shield Move		
SAP Part Number	Description	QTY
377929	BRACKET,BASELIFT CYLINDER	1
4140891	CABLE(HOSE-YEL)SKK24(Cr)/SKK24(Cr) 4.0M	50
351685	ISO ADAPTER PMC-R; II 1; SKK24	8
4140829	CABLE(HOSE-YEL)SKK24(Cr)/SKK24(Cr) 1.0M	5
4287723	SOLENOID DRIVER SD1 CR WITH PROT.CAP USA	25
318825	PLUG,CODE,10K OHMS	1
4394865	CABLE(HOSE) B66/OS4(Cr) 20FT	2
128743	STAPLE(LONG), 3/8" F/SKK24	100
165446	BAR (UV)	2
663369	PLUG-IN ELBOW LUMBERG-SKK24	3
315495	BLIND PLUG, SKK24 WITH O-RING	100
145336	ACCUMULATOR	2
310604	VALVE,NG20 PILOT OPRTD LEG CHK W/KN VLV	2
310603	VALVE,PILOT OPRTD LEG CHECK,W/OUT PLUGS	2
327868	VALVE,YIELD,420 BAR	3
4341364	HP-BLOCK BALL VALVE SIBK 2515/2S DN 25	3
204146	ELBOW,MANIFOLD,150 DEG DN40SUPERSTECKO	6
204144	ELBOW,MANIFOLD,150 DEGREE DN50	3
204141	ELBOW,MANIFOLD,150 DEGREE DN50	6
4367163	CABLE(BLUE)SKK24(Cr)/OS-4(Cr) 1:1 1800MM	14
4331307	CABLE(BLUE)SKK24(Cr)/OS-4(Cr) SD1 600MM	20
4331308	CABLE(BLUE)OS-4(Cr)/OS-4(Cr) SD1 1300MM	50
4224199	CABLE OS4 OS4 1800MM 1:1 CR	10
4367161	CABLE(BLUE) OS-4(Cr)/OS-4(Cr) SD1 1800MM	10

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6.0 Resources Required

(see implementation steps)

7.0 Related Best Practices

8.0 Acknowledgements

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